Himesh Ahuja

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Education

University of Virginia **August 2025 - May 2028**
Computer Science and Mathematics | GPA: 3.85 / 4.0 *Charlottesville, VA*

- **Honors:** Echols Scholar (Top 5% of Class), Teaching Assistant (Data Structures), Dean's List × 2
- **Relevant Coursework:** Advanced Algorithms, Computer Architecture, Machine Learning, LLM Engineering

Experience

Software Engineer Intern **January 2026 – Present**
Noblis *Reston, VA*

- Co-designed on-prem LLM platform with infrastructure engineers using vLLM, Kong, FastAPI, Docker, and Bash scripts, supporting 5 self-hosted open-weight models across 10+ microservices
- Refined request forwarding, reverse proxies, containerization, and Single Sign-On through senior engineer code reviews, productionizing secure API access and growing internal adoption 10× to 100+ engineers
- Implemented mutual TLS (mTLS) for service-to-service authentication and Kong request logging through DevSecOps design reviews, strengthening platform security and observability for 100K+ daily inference requests
- Configured vLLM deployment with systemd, achieving 99% uptime with Python polling and uptime dashboards
- Provisioned Linux development VMs, deployed test LLMs, and onboarded 4 engineers for concurrent development

Research Assistant **August 2025 – June 2026**
Chandra Robot Autonomy Lab *Charlottesville, VA*

- Engineered ROS2 (Foxy + RViz2) simulation and tuned Pure Pursuit (lookahead + PID) for autonomous-vehicle research, improving navigation success from 50% to ∼88% at <20 cm and earning a $2K Ingrassia Grant
- Tuned Pure Pursuit (lookahead + PID) under graduate mentorship, raising success from 50% to ∼88% at <20 cm
- Validated TurtleBot-to-Jackal navigation through 50+ trials with PhD researchers, enabling 75% higher speeds
- Migrated C++ launch files to Python, streamlining workflows by eliminating rebuilds and saving 30s per iteration

Software Engineer Intern **May 2024 – August 2024**
NT Concepts *Tysons Corner, VA*

- Collaborated with 8 engineers via agile sprints and code reviews to deliver GIS capabilities for Google subcontract
- Shipped 10+ GIS features in React, OpenLayers, Redux, including user location sharing and interactive map UI

Projects

reviewmyresumeplease.com | *Typescript, FastAPI, PostgreSQL, Docker Compose, Grafana* **Personal Project**

- Built resume review platform with PDF annotation, and revision tracking, across 100+ users and submissions
- Deployed Docker Compose stack with database and monitoring microservices, cutting setup to ∼1 minute
- Configured Nginx with TLS on NixOS VPS behind Cloudflare CDN; monitored on Grafana and Prometheus

theCourseForum | *Django, PostgreSQL, Docker, AWS, GitHub Actions* **Student Organization**

- Owned implementation of Django/PostgreSQL features for 20K+ students, collaborating in 20+ engineer team
- Reduced AWS pipeline runtimes by 47% (15+ to 8 minutes) via multistage Docker builds and dependency caching
- Architected threaded comment system by migrating the data model, supporting discussions on 200K+ reviews
- Implemented 15+ end-to-end tests to GitHub Actions, improving merge reliability and preventing regressions

A8THER | *Python, FastAPI, Docker, Ghidra, WebSockets, SQLite* **HooHacks 2026**

- Developed reverse engineering tool to decompile binaries, reducing analysis time to under 1 minute per file
- Orchestrated LLM reasoning over Ghidra, objdump, and readelf inside Docker sandbox; cached results in SQLite
- Integrated Xterm.js, linked to Docker sandbox via WebSocket, enabling command execution for users and agents

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL, x86-64 Assembly
Frameworks: FastAPI, React, Node.js, Flask, Django, LangChain, TensorFlow, PyTorch, Playwright, ROS2
Infrastructure: AWS, Docker, Kubernetes, Git, GitHub Actions, Redis, VPS, NixOS